Fifth Biweekly Report (June 1 – June 14, 2026)

Reporting Period: June 1 – June 14, 2026 (Weeks 12–13) | Submission: June 14, 2026

1. Team Summary — WBS Progress Overview

This final biweekly report covers Weeks 12–13 (June 1–14, 2026), during which HeartCompass completed all remaining work and delivered v1.0.0. Key achievements: WP3.6 prompt optimisation Phase 2 reached 100% with emotion-aware tone adaptation and cultural-context sensitivity (92% persona-consistency score across all 5 FigureRole types). WP4.6 dark-mode card palette finalised with WCAG AA contrast verification and cross-browser validation (100%). WP5.4 integration testing executed all 8 critical-path scenarios with zero failures (100%). WP5.6 final documentation, Harness assets, and deployment configuration completed (100%). All 30 WBS tasks are now at 100% completion. v1.0.0 tagged on June 13, 2026. Total cumulative effort: 298 person-hours. Risk: NONE — project complete.

1.1 WBS Task Progress Matrix (Weeks 12–13)

WBS Task	Description	Owner	Hours	Compl.	Status
WP3.6	Prompt Optimisation — Emotion-Aware Tone & Cultural Context	Yihan Liu	18	100%	Complete
WP4.6	Dark-Mode Card Palette & Cross-Browser Validation	Leishiyu Chen	10	100%	Complete
WP5.3	Docker Compose, pgvector Healthcheck & Alembic — Final	Haojie Hu	12	100%	Complete
WP5.4	Integration Test Execution (8 Critical-Path Scenarios)	Haojie Hu	18	100%	Complete
WP5.6	Final Documentation, Harness Assets & Deployment	Tian Bu (lead)	16	100%	Complete
WP2-5 (prior)	All Other Tasks — Persona Building, ConversationGraph Core, Lark Bot, CLI, PyPI (completed in prior periods)	Team	224	100%	Complete

1.2 Overall Assessment & Final Deliverables

The HeartCompass course project is complete. All charter-defined scope has been delivered: a five-layer architecture (Database → Service → Agent → API → Channel/CLI), two LangGraph StateGraph workflows (FRBuildingGraph, ConversationGraph), a Lark/Feishu Bot with 4-command menu and dark-mode cards, a comprehensive CLI (immortality: doctor, setup, auth, fr, logs, lark-service), Docker Compose with pgvector healthcheck (30 cold-start cycles, zero failures), Alembic migration pipeline, PyPI package (digital-immortality v1.0.0), and a reusable Harness code-quality workflow (Commit Quality Reviewer, automated change-log, Conventional Commits, global AI rules). The WP3 Conversation System achieved 92% persona-consistency with emotion-aware tone and cultural-context sensitivity; full regression testing passed 50/50 scenarios. The WP5.4 integration test suite passed 8/8 critical-path scenarios. All 12 bugs from the final sweep were resolved with zero high-risk issues remaining. Final effort distribution: WP1 30h (10.1%) / WP2 54h (18.1%) / WP3 80h (26.8%) / WP4 72h (24.2%) / WP5+Harness 62h (20.8%) = 298h total over 13 weeks.

Leishiyu Chen

RBS Area: Channel Integration & Card System (RBS R2.2.1, R2.2.2) | Total Hours: 24h

WBS Task	Description	Hours	Compl.	Status
WP4.6	Dark-Mode Card Palette — Finalisation & Cross-Browser	10	100%	Complete
WP4.5e	Lark Sandbox End-to-End Validation	6	100%	Complete
WP5.4c	Integration Test Support — Channel & WebSocket Scenarios	4	100%	Complete
WP5.6d	Lark Bot Deployment Documentation	4	100%	Complete

II. Period Summary

My work in Weeks 12–13 focused on finalising the WP4 Channel Integration pipeline and supporting the project's release activities. WP4.6 dark-mode card palette was completed to 100% with verified WCAG AA contrast ratios across all 4 card types, cross-browser rendering validation (Lark Desktop, Lark Mobile iOS/Android, Web), and null-safe fallback behaviour for every display field. WP4.5 Lark sandbox validation reached 100% after a full end-to-end exercise covering WebSocket connect/disconnect cycles, message replay on reconnect, and multi-device message synchronisation. I supported WP5.4 integration testing by executing and validating the 2 channel-specific scenarios (Lark WebSocket lifecycle and cross-platform card rendering), both passing on the first run. I also authored the Lark Bot deployment guide for the final documentation package. Total: 24h for this period.

III. Key Progress by WBS Task

[WP4.6 · 10h · 100%]

- Dark-Mode Card Palette — Finalisation & Cross-Browser Validation Completed the dark-mode palette for all 4 Lark interactive card types (persona display, build progress, conversation reply, error state). Verified WCAG AA contrast ratios (minimum 4.5:1 for normal text, 3:1 for large text) across all colour pairs. Cross-browser testing matrix: Lark Desktop (macOS 6.4, Windows 6.3), Lark Mobile (iOS 8.6, Android 8.5), and Lark Web — all passed. Null-safe rendering verified for 12 edge-case display fields (missing avatar, empty dimension values, truncated Chinese text, zero-count feeds). Final palette: navy #1F4E79 → slate #7890A0 for muted elements, teal #5D8AA8 for interactive states, pale #DDE9F2 for card backgrounds in dark mode.

[WP4.5e · 6h · 100%]

- Lark Sandbox End-to-End Validation Completed full end-to-end validation in the live Lark sandbox environment. Tested: WebSocket connect/disconnect cycles (30 iterations, zero failures), exponential-backoff reconnection (1s/2s/4s with max 3 retries, all recovered within 7s), message replay on reconnect (queued messages delivered in order, no duplicates), multi-device message synchronisation (desktop + mobile, 50 message pairs, zero inconsistencies), and 30s idle timeout with proactive reconnect. All scenarios passed. This closes the WP4 integration testing loop.

[WP5.4c / WP5.6d · 8h · 100%]

- Integration Test & Documentation Support Executed the 2 channel-specific integration test scenarios for WP5.4: (1) Lark WebSocket lifecycle — connect, authenticate, receive message, dispatch to ConversationGraph, receive reply, deliver to user, disconnect/reconnect with message replay; (2) cross-platform card rendering — verify all 4 card types render correctly on Desktop, Mobile, and Web with both light and dark palettes. Both scenarios passed on the first run. Authored the Lark Bot Deployment Guide (Section 4 of final documentation): setup instructions, credential configuration, WebSocket endpoint registration, and troubleshooting for common sandbox issues.

Yihan Liu

RBS Area: Conversation & Dialogue System (RBS R2.1.1, R2.1.2) | Total Hours: 28h

WBS Task	Description	Hours	Compl.	Status
WP3.6	Prompt Optimisation Phase 2 — Emotion-Aware Tone & Culture	16	100%	Complete
WP3.5f	Full Conversation Regression Testing (50 Scenarios)	8	100%	Complete
WP5.6p	Prompt Engineering Documentation & Template Catalog	4	100%	Complete

II. Period Summary

My work in Weeks 12–13 completed the WP3 Conversation System to its final production quality. WP3.6 Phase 2 delivered emotion-aware tone adaptation and cultural-context sensitivity, building on Phase 1's anti-repetition and formality tuning. The final prompt system achieved 92% persona-consistency score across all 5 FigureRole types (family, friend, mentor, colleague, partner), up from 78% before Phase 2. I implemented emotion detection from conversation context (5-class: joy, sadness, anger, anxiety, neutral) with corresponding tone parameters, and cultural-context markers for Chinese-language interaction norms (indirectness preference, seniority-aware address forms, high-context implication handling). A comprehensive 50-scenario regression suite was executed across all FigureRole types, covering standard dialogue, rapid topic switching, 500+ word monologues, Chinese/English code-switching, and edge cases — achieving 100% pass rate. I also compiled the prompt engineering documentation cataloguing all 12 prompt templates with their design rationale, parameter schemas, and ablation results. Total: 28h for this period.

III. Key Progress by WBS Task

[WP3.6 · 16h · 100%]

▸ Prompt Optimisation Phase 2 — Emotion-Aware Tone & Cultural Context Completed the second and final phase of prompt optimisation. Emotion-aware tone adaptation: integrated 5-class emotion detection (joy/sadness/anger/anxiety/neutral) from the last 3 conversation turns, mapping each emotion to tone parameters (warmth 0.2–0.9, formality 0.1–0.7, verbosity 0.3–0.8, empathy 0.2–0.95). Cultural-context sensitivity: added Chinese-language interaction norms including indirectness preference (hedging for sensitive topics), seniority-aware address forms (您 vs 你 based on age/role differential), and high-context implication handling (recognising unstated concerns from contextual cues). Quantitative results: persona-consistency score 92% (up from 78% pre-Phase-2), emotion-appropriateness rating 4.2/5.0 in blind human evaluation (n=150 responses across 5 FigureRole types), cultural-appropriateness rating 4.4/5.0. Phase 2 required 16h due to iterative prompt refinement and ablation testing across 12 prompt variants.

[WP3.5f · 8h · 100%]

▸ Full Conversation Regression Testing — 50 Scenarios, 100% Pass Designed and executed a comprehensive 50-scenario regression test suite covering all 5 FigureRole types (10 scenarios each). Scenario categories: standard multi-turn dialogue (20), rapid topic switching (8), 500+ word monologue handling (6), Chinese/English code-switching (6), edge cases including empty recall, conflicting feeds, and maximum memory trim (10). All 50 scenarios achieved 100% pass rate. Key metrics validated: response relevance ≥ 0.85, persona consistency ≥ 0.90, context retention across ≥ 10 turns, memory trim correctness (no context bleed between unrelated topics). Automated test harness with JSON scenario definitions enables reproducible regression testing for future maintenance.

[WP5.6p · 4h · 100%]

▸ Prompt Engineering Documentation & Template Catalog Compiled comprehensive prompt engineering documentation for the final project deliverable. Catalogued all 12 prompt templates (4 FRBuildingGraph extraction prompts × 4 dimensions + 5 ConversationGraph role-specific system prompts + 3 utility prompts for memory trim, feed sync, and conflict detection). Each template entry includes: design rationale, parameter schema with type constraints, ablation results comparing 2–3 variants with quantitative metrics, and usage guidelines for future contributors. This documentation supports the project's reproducibility and maintainability requirements for the final submission.

Haojie Hu

RBS Area: Infrastructure, Vector Recall & Deployment (RBS R1.2.2, R2.1.1, R3.1.2) | Total Hours: 24h

WBS Task	Description	Hours	Compl.	Status
WP5.4	Integration Test Execution — 8 Critical-Path Scenarios	14	100%	Complete
WP5.3	Docker Compose, pgvector Healthcheck & Alembic — Final	4	100%	Complete
WP5.6	Deployment Configuration & v1.0.0 Release Build	6	100%	Complete

II. Period Summary

My work in Weeks 12–13 centred on system integration testing, deployment finalisation, and the v1.0.0 release. WP5.4 integration testing was the primary effort: I executed all 8 critical-path scenarios covering CLI-to-DB connectivity, Lark WebSocket lifecycle, FRBuildingGraph end-to-end, ConversationGraph multi-turn coherence, Docker Compose cold-start, pgvector healthcheck, auth token lifecycle, and cross-platform card rendering. All 8 scenarios passed with zero failures; results were documented with execution logs, timing measurements, and resource utilisation data. WP5.3 Docker deployment was finalised to 100%: the custom pgvector healthcheck (pg_isready + vector type availability probe) passed 30 consecutive cold-start cycles, and Alembic migration 003 was validated with forward/rollback/re-forward cycle testing. For WP5.6, I configured the final deployment environment, built the v1.0.0 release artefacts, and verified the PyPI package (digital-immortality v1.0.0) with a clean installation test. Total: 24h for this period.

III. Key Progress by WBS Task

[WP5.4 · 14h · 100%]

▸ Integration Test Execution — 8/8 Scenarios Passed Executed the full integration test plan approved in Week 11. All 8 critical-path scenarios passed with zero failures: (1) CLI-to-DB Connectivity — immortality doctor validates PostgreSQL + pgvector, setup creates tables, FR add/list/show operate correctly; (2) Lark WebSocket Lifecycle — connect, authenticate with app_id/secret, receive message event, dispatch to ConversationGraph, receive reply, deliver to user, disconnect with graceful shutdown; (3) FRBuildingGraph End-to-End — submit source material, verify preprocessing, confirm FineGrainedFeed extraction across 4 dimensions, validate FR core field updates, check FROverallUpdateLog; (4) ConversationGraph Multi-Turn Coherence — 20-turn dialogue across 3 FigureRole types, verify persona consistency, context retention, and memory trim correctness; (5) Docker Compose Cold-Start — docker compose up, verify pgvector healthcheck, Alembic migration auto-run, app readiness; (6) pgvector Healthcheck — verify custom SQL probe (pg_isready + vector type in pg_available_extensions), 30-cycle validation; (7) Auth Token Lifecycle — login → token persistence → authenticated request → token expiry → 401 → transparent refresh → retry success → logout → token revocation; (8) Cross-Platform Card Rendering — verify 4 card types on Desktop/Mobile/Web, light and dark palettes. All scenarios documented with execution logs and timing data.

[WP5.3 / WP5.6 · 10h · 100%]

▸ Deployment Finalisation & v1.0.0 Release Finalised Docker deployment: validated custom pgvector healthcheck script across 30 consecutive cold-start cycles (docker compose down -v && docker compose up), mean startup time 8.3s, zero failures. Alembic migration 003 verified with forward/rollback/re-forward cycle — all schema changes applied and reverted cleanly. Built v1.0.0 release artefacts: tagged on GitHub (June 13, 2026), verified PyPI package digital-immortality v1.0.0 with clean pip install in a fresh venv, confirmed all CLI commands functional post-install. Deployment documentation updated with Docker Compose quickstart, environment variable reference, and troubleshooting guide for common issues (port conflicts, pgvector extension not available, API key misconfiguration).

Tian Bu (Team Leader)

RBS Area: CLI, Project Governance & Delivery (RBS R3.1.1, R3.2) | Total Hours: 30h

WBS Task	Description	Hours	Compl.	Status
WP5.6	Final Documentation — README, API Docs, Architecture Diagram, Deployment Guide	12	100%	Complete
A4	Bugfix Sweep & Regression Validation	8	100%	Complete
A6	Milestone 3 Final Deliverable & v1.0.0 Release Management	6	100%	Complete
—	Project Governance & Fifth Biweekly Report	4	—	Complete

II. Period Summary

My work in Weeks 12–13 focused on bringing the project to its final delivery state. WP5.6 final documentation was the largest effort: I overhauled the project README with the final five-layer architecture diagram, coordinated the API documentation, CLI usage guide, Lark Bot setup instructions, and Docker Compose deployment guide into a cohesive documentation package. The bugfix sweep (Action A4 from the Milestone 2 review) systematically addressed 12 issues identified by the Commit Quality Reviewer and manual testing — all resolved with zero high-risk issues remaining. I managed the v1.0.0 release process (A6): coordinated the final merge, verified all 30 WBS tasks at 100%, tagged the release, and assembled the final project submission package including source code, documentation, WBS tracking summary, integration test results, and Milestone 2 review meeting summary. I also maintained WBS tracking through Weeks 12–13, coordinated the final team review, and compiled this fifth and final biweekly report. Total: 30h for this period.

III. Key Progress by WBS Task

[WP5.6 · 12h · 100%]

‣ Final Documentation — Complete Project Documentation Package Compiled the complete project documentation package as the capstone of WP5.6. README overhaul: added final five-layer architecture diagram (ASCII art + textual description), project overview with scope and objectives, quickstart guide covering CLI setup + Lark Bot configuration + Docker deployment in under 10 minutes, and links to all subsidiary documents. API documentation: documented all FastAPI endpoints with request/response schemas, authentication requirements, and example curl commands. CLI usage guide: complete reference for all 7 immortality commands (doctor, setup, auth login/logout/whoami, fr add/list/show, logs, lark-service) with option tables, exit codes, and JSON output examples. Lark Bot setup instructions: credential acquisition, WebSocket endpoint registration, bot command reference, and sandbox testing procedure. Docker Compose deployment guide: prerequisites, step-by-step startup, healthcheck verification, and troubleshooting. All documents written in English, formatted for GitHub Markdown rendering.

[A4 / A6 · 14h · 100%]

‣ Bugfix Sweep & v1.0.0 Release Management Bugfix sweep (A4): systematically triaged and resolved 12 issues — 3 from Commit Quality Reviewer findings (2 medium-risk: session file edge case on NFS filesystems, card template text truncation for 4-byte UTF-8 characters; 1 low-risk: log level default inconsistency), 5 from manual testing (FR field partial update edge cases, Lark message dedup false positive for identical subsequent messages within 15s, conversation memory trim off-by-one at exact boundary, pgvector healthcheck race condition on slow disks, CLI setup non-interactive mode fallback), and 4 cosmetic fixes (typos, inconsistent naming, missing help text). Post-fix regression suite re-run: all 50 conversation scenarios + all 8 integration scenarios passed. v1.0.0 release (A6): coordinated final main-branch merge (squash-merged 3 feature branches), verified all 30 WBS tasks at 100%, tagged v1.0.0 on June 13, 2026, assembled final submission package (source tarball, PDF documentation set, WBS tracking spreadsheet, integration test report, meeting summaries).

[Governance · 4h]

‣ Project Governance & Fifth Biweekly Report Maintained WBS tracking through the final sprint, updating per-task completion percentages and actual hours as tasks closed. Coordinated the final team review meeting (June 12) to verify all deliverables against the Project Charter success criteria. Compiled this fifth biweekly report from individual member submissions, ensuring consistency with the WBS tracking data and the Project Charter's planned scope. All figures cross-validated: cumulative 298h across 4 members, 30/30 tasks at 100%, v1.0.0 delivered on schedule.